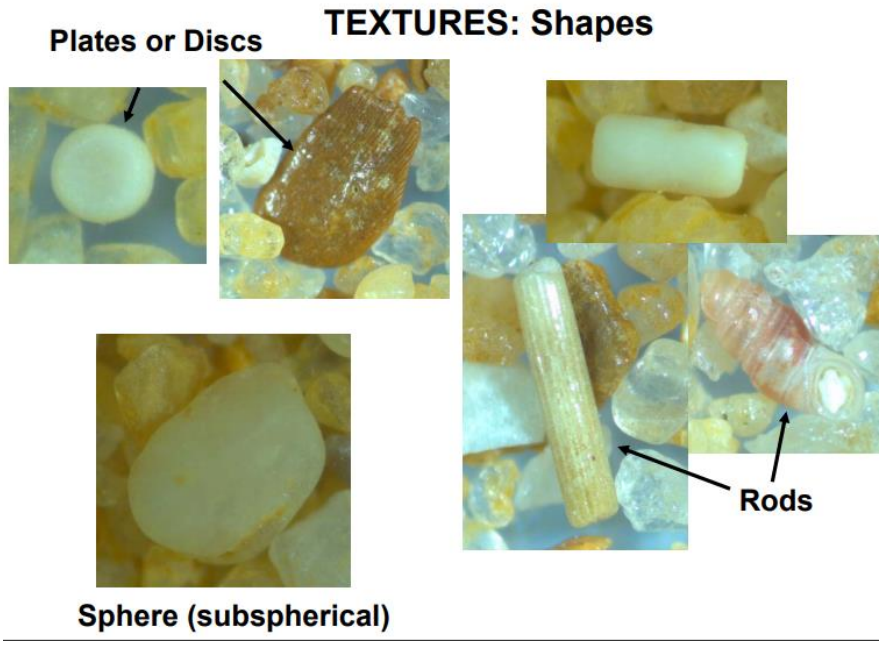
Describing the physical characteristics of sediment

Grain shape

We want to note two things about the grain shape: the angularity (usually, and misleadingly, called the roundness) and the sphericity.

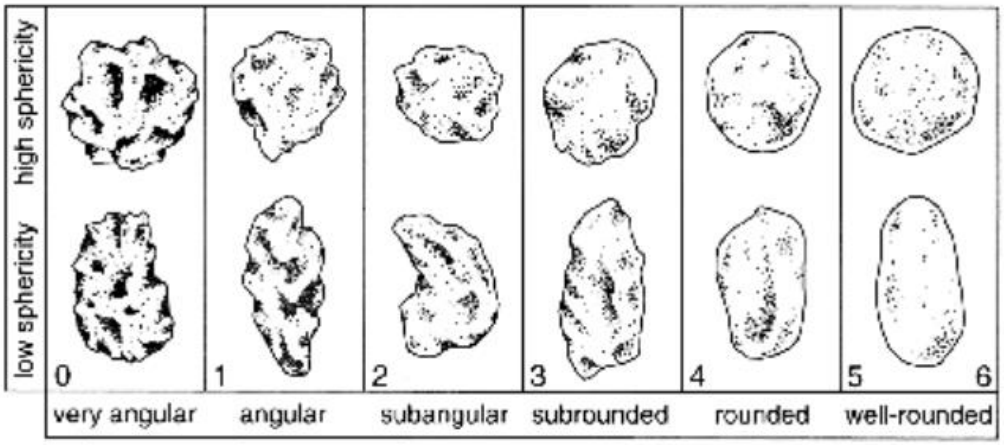
Angularity/roundness refers to whether the grains are smooth or jagged, and we describe grains as well or poorly rounded. **Sphericity** refers to how equant their overall shape is determined by the ratio of the long, intermediate and short axes of the grain. if they aren't spherical we describe them using standard geometric terms (e.g. prolate spheroid).



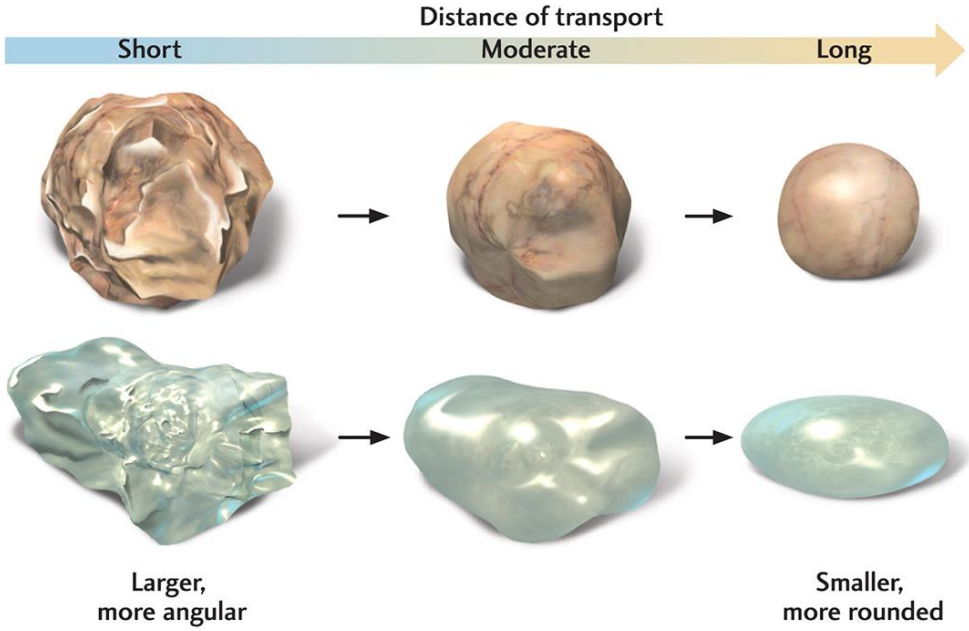
The four classes of grain shape: spheres, discs, rods and blades (after Illenberger, 1991)

In practice we usually just need to estimate roundness and sphericity using simple comparators such as the one shown below. A very useful one is found on your grain size card which you have in your field kit.

Categories of roundness for sediment grains (after Pettijohn et al., 1987)



ure 9 Categories of roundness for sediment grains (after Pettijohn et al., 1987)



Grotzinger/Jordan, *Understanding Earth*, 8e, © 2020 W. H. Freeman and Company

Interpretation

The angularity of sediment grains is useful in that it tells us something about how far and how vigorously the grains have been transported. So, for example, a population of very well rounded grains does suggest either many cycles of transport or intense abrasion in a high energy environment (e.g. a desert or beach).

This is mostly useful in sand-sized or larger sediment; the finer the grains the more they are, in effect, cushioned from impacts by the fluid they travel in. So angular silt isn't anything to write home about. But large grains generally get their corners knocked off pretty quickly— in a few km or tens of km for gravel, typically.

Sphericity is less useful for interpretation: the gross shape of grains mainly reflects the anisotropy of the parent material (i.e. the composition of the parent material or any planes of weakness in the rock e.g. bedding/lamination, cleavage or jointing). Rod-shaped blobs (e.g. veins) tend to weather to football-shaped clasts, while slates produce disks, as you would expect.